

KUO-HAN HUNG

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Education

Stanford University

June 2027 (Expected)

M.S. in Computer Science

Stanford, CA

- GPA: 4.0/4.0; Specialization: Artificial Intelligence and Robotics.

National Taiwan University

Dec. 2024

B.S. in Computer Science and Information Engineering

Taipei, Taiwan

- GPA: 4.24/4.3.
- Awards: Bachelor Thesis Dean's Award, Presidential Award (×2), Dean's List (×3).

Technical Skills

- **Languages:** Python, C/C++, Golang, JavaScript, SQL, Bash.
- **Research Areas:** Reinforcement Learning, Vision-Language-Action Models, LLMs & VLMs, Diffusion Models, Mechanistic Interpretability.
- **ML Toolkits:** PyTorch, JAX, NumPy, Scikit-Learn, Pandas, Matplotlib.
- **Infrastructure & Other:** Linux, Docker, Git, ROS, Azure, React, MongoDB.

Research Experience

Stanford Artificial Intelligence Laboratory (ILIAD & IRIS Labs)

Sept. 2025 – Present

Graduate Researcher, Advisors: Prof. Dorsa Sadigh & Prof. Chelsea Finn

Stanford, CA

- **RL finetuning of real-time VLA policies** under inference latency: raised success rate from 42% to 97% on four dynamic real-world tasks. [1]
- **Sample-efficient RL finetuning of pretrained VLAs:** 30/30 success on all tasks with 19 min of online data on average. [2]
- **Scaling Q-functions** with Transformers in offline RL: 43% gain as model size grows. [3]
- **Visual representation pretraining** for reinforcement learning (ongoing).

Robot Learning Lab

Oct. 2024 – May 2025

Undergraduate Researcher, Advisor: Prof. Shao-Hua Sun

Taipei, Taiwan

- **Manipulation policies learned from action-free videos**, using optical flow as an action surrogate. [5]

Communication and Multimedia Lab

May 2022 – May 2024

Undergraduate Researcher, Advisors: Prof. Winston H. Hsu & Prof. Yi-Ting Chen

Taipei, Taiwan

- **Vision-instruction correlation rewards** for long-horizon manipulation: 43% improvement over state-of-the-art baselines. [4]
- **Behavior error detection in few-shot imitation policies** via representation pattern analysis.

Work Experience

IBM Research

June 2024 – Sept. 2024

Research Intern, Host: Dr. Pin-Yu Chen

Yorktown Heights, NY

- Found that attention drifts from the original instruction during **prompt injection attacks on LLM agents**.
- Traced the effect to specific attention heads using mechanistic interpretability.
- Designed an attention-based detector, a 31% AUROC improvement over the prior state of the art.

Microsoft

July 2023 – June 2024

Applied Scientist Intern, Host: Dr. Chiquan Zhang

Mountain View, CA (Remote)

- **Integrated Bing Maps with Microsoft Copilot** for conversational, multi-destination map queries.
- Formulated a pathfinding algorithm that learns from history, **achieving a 10× speedup over Dijkstra**.
- Developed an LLM-powered search system that adapts to user intent and geographic context.

Selected Publications

(* indicates equal contribution)

- [1] **Reinforcement Learning for Real-Time Vision-Language-Action Policies**
Perry Dong*, **Kuo-Han Hung***, Dorsa Sadigh, and Chelsea Finn
Under Review
- [2] **EXPO-FT: Sample-Efficient Reinforcement Learning Finetuning for Vision-Language-Action Models**
Perry Dong*, **Kuo-Han Hung***, Tian Gao, Dorsa Sadigh, and Chelsea Finn
Conference on Robot Learning (CoRL), 2026
- [3] **TQL: Scaling Q-Functions with Transformers by Preventing Attention Collapse**
Perry Dong*, **Kuo-Han Hung***, Alexander Swerdlow, Dorsa Sadigh, and Chelsea Finn
International Conference on Machine Learning (ICML), 2026
- [4] **VICtoR: Learning Hierarchical Vision-Instruction Rewards for Long-horizon Manipulation**
Kuo-Han Hung*, Pang-Chi Lo*, Jia-Fong Yeh*, Han-Yuan Hsu, Yi-Ting Chen, and Winston Hsu
International Conference on Learning Representations (ICLR), 2025
- [5] **Learning Skill Abstraction from Action-Free Videos via Optical Flow**
Hung-Chieh Fang*, **Kuo-Han Hung***, Chu-Rong Chen, . . . , Min-Hung Chen, and Shao-Hua Sun
ICML Workshop on Building Physically Plausible World Models, 2025